

Contribution Report

Ensemble Methods: Boosting vs. Bagging
AI Academy — Machine Learning Final Project, Spring 2026
Team Sigmoid

Team Members

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Emin Huseynli	ehuseynli96-ops	ehuseynli96@gmail.com
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Individual Contributions

Member	Modules / Code	Report Sections	Experiments	Slides	Other
Emin Huseynli (ehuseynli96-ops)	src/trees/decision_tree.py src/boosting/adaboost.py src/boosting/samme_r.py experiments/run_all.py experiments/bonus_samme_r.py	§2 Decision Tree §3 AdaBoost	Exp. 1, Exp. 2	Methods (Tree / AdaBoost)	Repo setup, .gitignore, initial scaffolding
Ziyad Muradov (thedarkstringg)	src/bagging/random_forest.py src/boosting/one_vs_rest_adaboost.py src/utils/preprocessing.py src/utils/datasets.py src/metrics/evaluation.py	§4 Random Forest §6 Exp. Setup	Exp. 3, Exp. 4, Exp. 5	Results (RF / CV tables)	requirements. txt, dataset acquisition
Ismayil Yusifli (ismayilysfli)	src/unsupervised/pca.py src/unsupervised/kmeans.py src/unsupervised/dbscan.py experiments/*.py (1–7) experiments/bonus_tsne.py experiments/common.py	§1 Intro §7 Unsup. Conclusion	Exp. 6, Exp. 7	Full deck assembly, intro/ conclusion	Report LaTeX assembly, figures pipeline

Sign-off

All three members reviewed and approved the final codebase, report, and experimental results via GitHub pull request review. By signing below, each member confirms that the contributions listed above accurately reflect their individual work on this project.

Emin Huseynli

Ziyad Muradov

Ismayil Yusifli